

NeurIPS 2022 | When Missing Data Is a Signal: Making Clinical Survival Models Robust

DeepJoint, from the University of Cambridge and the University of Manchester, models how clinical data are observed, not just their values, and stays accurate when hospital practice shifts.

A blood test ordered at 3 a.m., a lab panel repeated every hour, a measurement that simply never appears: to the clinician at the bedside, none of these is a random event. They reflect a judgment about who is sick and who needs watching. Yet most survival models trained on electronic health records treat the timing and the absence of measurements as noise, to be imputed away or ignored, and learn only from the recorded values.

The pattern of when and which tests get ordered, what the authors call clinical presence, is itself informative, and it is not stable. It differs between hospitals, drifts over the years, and changes even between a weekday and a weekend admission. A model that quietly learns one hospital's ordering habits can degrade when it is deployed somewhere those habits differ, which is exactly the situation any model faces the moment it leaves the dataset it was trained on.

DeepJoint (NeurIPS 2022), from the MRC Biostatistics Unit at the University of Cambridge and the Health e-Research Centre at the University of Manchester, asks what happens if we model the observation process explicitly instead of ignoring it. It is a single multi-task recurrent network that predicts three dimensions of clinical presence jointly with the survival outcome. The paper shows this both sharpens short-term predictions and, more importantly, makes them markedly more robust when clinical practice shifts.

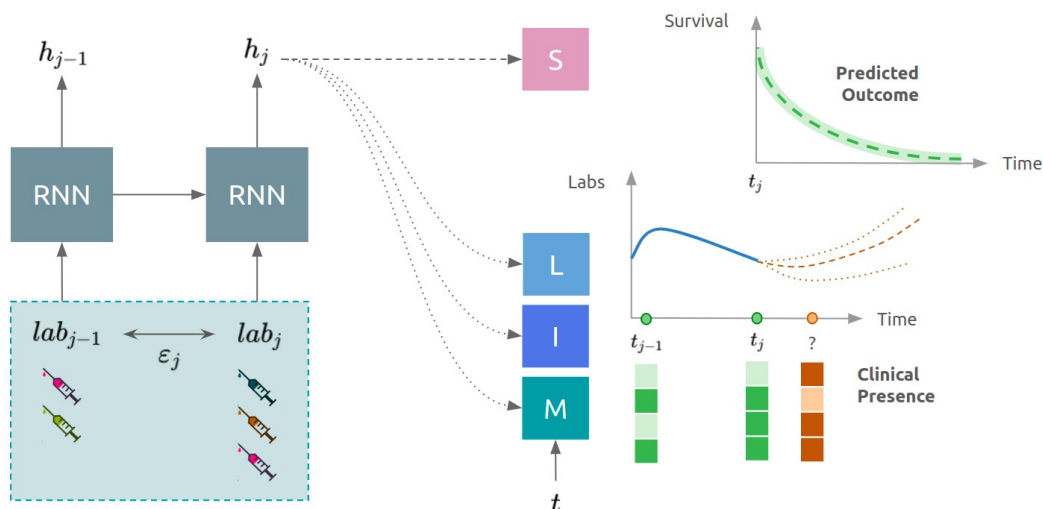


Figure 1. The Deep Joint Model. An LSTM turns a patient's irregular sequence of lab tests into an embedding h , which then feeds four heads: **L** models the longitudinal evolution of the values, **I** the inter-observation timing, **M** the missingness pattern, and **S** the survival outcome. Three of the four heads describe how the data are observed, not what they measure.

Paper: <https://arxiv.org/abs/2205.13481>

Code: <https://github.com/JeanSelme/MultitaskTransportability>

Clinical presence: the observation process as signal

Machine learning has studied distribution shift for years, mostly as covariate shift (the inputs change) or label shift (the outcome balance changes). Clinical presence is a third, less-examined kind of shift: the process that generates the observations changes. The same underlying patient can look very different in the data depending on how closely they were monitored, and hospitals monitor differently.

The usual response is to make the sampling process disappear. Impute the missing values, resample onto a regular grid, and hand the model a clean matrix. DeepJoint takes the opposite view: the missingness, the timing, and the trajectory of the measurements carry information about the patient's state, so the model should learn them rather than erase them. The bet is that a representation that understands the observation process will both predict better and travel better.

One embedding, four heads

At the core is a Long Short-Term Memory (LSTM) network that reads a patient's irregular sequence of laboratory tests after ICU admission and compresses it into an embedding h . That single embedding then drives four heads, each a small network with its own likelihood. The longitudinal head L predicts the next test values under a Gaussian likelihood; the missingness head M predicts which tests appear next under a Bernoulli likelihood; the inter-observation head I models the timing between measurements through a cumulative intensity; and the survival head S is a DeepSurv Cox model predicting the outcome.

All four heads are trained together. The survival loss and the three clinical-presence losses are combined into one objective, balanced by a hyperparameter α and a dynamic weighting-average scheme that keeps any single task from dominating: roughly, the loss is $(1 - \alpha)$ times the survival term plus α times a weighted sum of the longitudinal, timing, and missingness terms. Because the heads share h , forcing the embedding to explain how the data were observed regularizes it, which is what later pays off under shift.

The paper studies three variants of this idea. DeepJoint uses only the shared embedding for survival. DeepJointFeature adds explicit clinical-presence features on top of the embedding. DeepJointFineTune first trains the joint model, then fine-tunes the survival head end-to-end for maximum discrimination. As the results show, these three sit at different points on a trade-off between raw accuracy and robustness.

The data: 30,834 ICU patients from MIMIC-III

Experiments use MIMIC-III, the intensive-care database of anonymized labs, vitals, and diagnoses from Beth Israel Deaconess Medical Center covering 2001 to 2012. Of its 38,597 patients, the study keeps the 30,834 who survived at least the first 24 hours after admission, and predicts in-hospital survival from the embedding taken at the last observation within that

first day. Models are compared on a 90% / 10% patient-level split using the time-dependent concordance index (C-index, higher is better) and the Brier score, at horizons of 1, 7, and 14 days.

Modelling the observation process pays off, on short horizons

On a random split of the population, the payoff is clearest at the one-day horizon. Read the figure below by row: each model's C-index at 1, 7, and 14 days, with 95% confidence intervals; the proposed methods are in bold. The telling comparison is at 1 day. DeepJoint, which sees only the laboratory values, reaches 0.862, ahead of the LSTM that ignores clinical presence (Ignore, 0.853) and of GRU-D (0.855), a strong baseline that explicitly consumes missingness as an input. Modelling the observation process, even without feeding it in, produces a more predictive embedding.

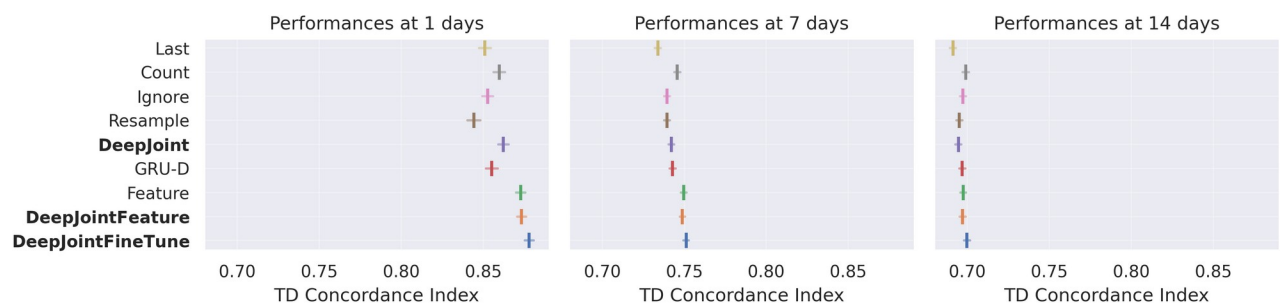


Figure 2. Time-dependent C-index on a random split, at 1, 7, and 14 days (95% confidence intervals; higher is better). The proposed methods (bold) lead on the short horizon and converge with the baselines as the horizon lengthens, consistent with clinical presence signalling short-term instability.

Adding features and fine-tuning pushes further: DeepJointFeature reaches 0.873 and DeepJointFineTune 0.878 at one day, the best in the table. But the advantage is horizon-dependent. By 7 and 14 days the models bunch together, because clinical presence mostly signals short-term instability, how closely a patient is being watched right now, rather than their long-run trajectory. The table gives the exact numbers.

Table 1. Time-dependent C-index (mean, higher is better) on a random population split. Proposed methods marked. The short-horizon edge is real but narrows by day 14.

Model	1 day	7 days	14 days
Last	0.851	0.734	0.691
Count	0.860	0.746	0.699
Ignore	0.853	0.740	0.697
Resample	0.844	0.739	0.695
GRU-D	0.855	0.743	0.697
Feature	0.873	0.750	0.697
DeepJoint (ours)	0.862	0.742	0.695
DeepJointFeature (ours)	0.873	0.749	0.697
DeepJointFineTune (ours)	0.878	0.751	0.700

The real test: surviving a shift in practice

Discrimination on a random split is the easy case, where train and test share the same observation process. The paper's central claim is about robustness, so it needs a controlled shift. It uses the weekend effect: patients admitted on weekends are monitored differently and are known to have different mortality from weekday admissions. That makes weekday-versus-weekend a natural, controllable change in the observation process.

Because the two groups also differ in outcome, their scores cannot be compared head to head. The paper instead borrows the robustness literature's transfer setup, shown below: train a model on one group and test it on the other, and separately train an oracle on the target group itself. Comparing the transferred model against its oracle isolates how much performance is lost purely to the shift in clinical presence.



Figure 3. The robustness protocol. Model 1 trains on weekend admissions and is tested on weekday admissions; model 2 trains and tests within weekday admissions, serving as the oracle for that target group. The gap between them measures sensitivity to the shift, not to the outcome difference.

The result is read against the diagonal. In the figure below, each model's transferred performance (y-axis) is plotted against its oracle performance (x-axis) at all three horizons; a point on the dotted diagonal means the shift cost nothing. Points that fall below the line have lost accuracy to the change in observation process, and the further below, the less robust the model.

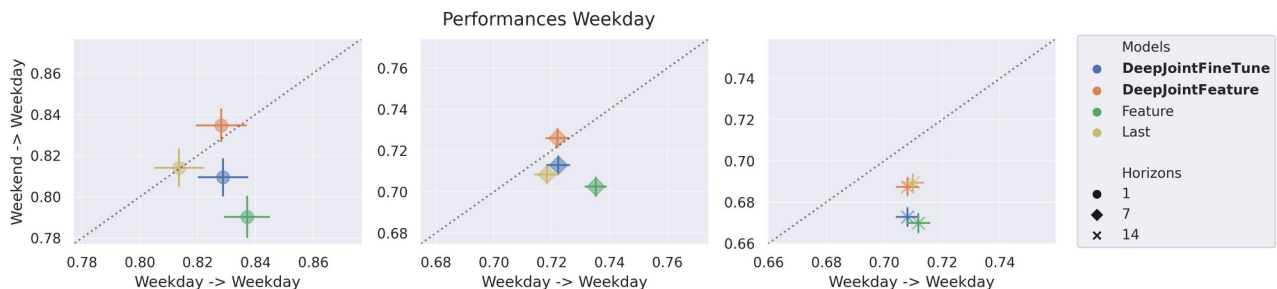


Figure 4. Robustness under the weekday-to-weekend shift: transferred C-index (y-axis) versus oracle C-index (x-axis) across horizons; closer to the diagonal is more robust. DeepJointFeature (orange) sits nearest the diagonal while keeping strong discrimination; DeepJointFineTune (blue), the most accurate model on the random split, drops further below it.

Here the ranking flips. DeepJointFineTune, the sharpest model on the random split, sits further below the diagonal: fine-tuning hard for discrimination made it lean on the training group's specific ordering habits, so it overfits the observation process. DeepJointFeature stays closest to the diagonal while still discriminating well, making it the best all-round choice under

shift. The accuracy champion and the robustness champion are not the same model, and the paper argues that for deployment the robust one is what matters.

Clinical presence is signal, not noise

The through-line is simple: how clinical data are observed, not just their recorded values, is information, and a model that learns the observation process alongside the outcome is both more accurate on short horizons and more transportable when practice changes. That reframes missingness and irregular timing from a preprocessing nuisance into something worth modelling directly.

The boundaries are worth stating plainly. The predictive edge is concentrated at short horizons and fades by two weeks; the study is a single database (MIMIC-III) and a single outcome (in-hospital survival); and the sharpest variant is also the one most prone to overfitting the observation process, so robustness has to be chosen for, not assumed. Still, the core message travels well beyond this dataset: when a model will be deployed somewhere its training habits do not hold, treating the observation process as signal is what lets its predictions survive the move.